

Logic - Well Founded Induction

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1 Induction on $\mathbb{N}$

1.1 What is $\mathbb{N}$?

Definition 1.1 (Peano-Like). *A structure $(N, S_N, 0_N)$ is called **Peano-Like** when*

(PL0) N is a set

(PL1) $0_N \in N$

(PL2) $S_N : N \rightarrow N$

(PL3) S_N is injective

(PL4) $\forall n \in N. S_N(n) \neq 0_N$

(PL5) $\forall n \in N. n = 0_N \vee \exists k \in N. (S_N(k) = n)$

We write $n + 1 := S_N(n)$, and for $n \in N$ we write ${}_S[n] := \{k \in N \mid S_N(k) = n\}$ for the set of predecessors of n .

Axiom 1.2. *There exists a Peano-Like structure. We fix one and call it $(\mathbb{N}, S, 0)$.*

Nothing in (PL0) to (PL5) says that induction holds. The conatural numbers $\mathbb{N} \cup \{\infty\}$ with $S(\infty) = \infty$ satisfy all six conditions and violate both propositions below. Everything in this section is therefore proved for an arbitrary Peano-Like structure, and $\mathbb{N}$ can be read as any one of them.

Remark (on the choice of axioms). The proof of Theorem 1.5 below uses (PL5) in the $\rightarrow$ direction, where we split n into the case $n = 0$ and the case $n = S(k)$. There were other ways to set things up.

Option 1: drop (PL5). Then the equivalence is false, not just harder to prove. Let X be two disjoint copies of the natural numbers, with S the usual successor on each copy and 0 the bottom of the first copy. Then (PL0) to (PL4) hold, but the bottom c of the second copy is not 0 and has no predecessor. Proposition 1.3 fails: take B to be the

first copy, which contains 0 and is closed under S but is not all of X . Proposition 1.4 holds: any B satisfying its hypothesis contains 0 and c , since both have $s[\cdot] = \emptyset$, and then contains every successor of an element of B , hence both copies. So Proposition 1.4 holds and Proposition 1.3 fails in this structure. The same B shows that the pointwise equivalence used in the proof below fails, since $s[c] = \emptyset \subseteq B$ while $c \notin B$.

Option 2: use the usual Peano axioms with induction as an axiom. Take (PL0) to (PL4) together with Proposition 1.3. Then (PL5) becomes a lemma, by applying induction to $\{0\} \cup S[\mathbb{N}]$, which contains 0 and is closed under S . This setup makes the equivalence less informative, since the induction principle is assumed instead of derived from the structure. (PL3) and (PL4) are still needed, but in the $\leftarrow$ direction of the proof below.

My choice. I keep (PL5) as an axiom of the structure. This isolates what the equivalence really depends on, and it allows the direct proof by reducing the goal for each fixed B .

1.2 The induction statement on $\mathbb{N}$

Proposition 1.3. *For every $B \subseteq \mathbb{N}$*

$$[0 \in B \wedge \forall n \in \mathbb{N}. (n \in B \rightarrow n + 1 \in B)] \rightarrow B = \mathbb{N}$$

Proposition 1.4. *For every $B \subseteq \mathbb{N}$*

$$[\forall n \in \mathbb{N}. (s[n] \subseteq B \rightarrow n \in B)] \rightarrow B = \mathbb{N}$$

Theorem 1.5.

$$\text{Prop 1.3} \longleftrightarrow \text{Prop 1.4}$$

Lemma 1.6 (Reduction). *Let X be a set and $\varphi_1, \varphi_2, \psi$ be predicates on X with $\forall x \in X. (\varphi_1(x) \longleftrightarrow \varphi_2(x))$. Then*

$$\forall x \in X. (\varphi_1(x) \rightarrow \psi(x)) \longleftrightarrow \forall x \in X. (\varphi_2(x) \rightarrow \psi(x)).$$

Proof. For $\rightarrow$, assume the left side and take x with $\varphi_2(x)$. Then $\varphi_1(x)$ by the assumed equivalence, so $\psi(x)$. The $\leftarrow$ direction is the same with φ_1 and φ_2 exchanged. $\square$

Proof of Theorem 1.5. We apply Lemma 1.6 with $X = \mathcal{P}(\mathbb{N})$, $x = B$, $\psi(B) := (B = \mathbb{N})$,

$$\varphi_1(B) := 0 \in B \wedge \forall n \in \mathbb{N}. (n \in B \rightarrow n + 1 \in B),$$

$$\varphi_2(B) := \forall n \in \mathbb{N}. (s[n] \subseteq B \rightarrow n \in B).$$

Since ψ is the same on both sides, it is enough to show $\varphi_1(B) \longleftrightarrow \varphi_2(B)$ for every $B \subseteq \mathbb{N}$.

Take arbitrary $B \subseteq \mathbb{N}$.

Firstly, $\varphi_1(B) \rightarrow \varphi_2(B)$. Assume

$$0 \in B \wedge \forall n \in \mathbb{N}. (n \in B \rightarrow n+1 \in B). \quad (1)$$

W.T.S¹

$$\forall n \in \mathbb{N}. ({}_S[n] \subseteq B \rightarrow n \in B).$$

Take arbitrary $n \in \mathbb{N}$ with ${}_S[n] \subseteq B$. We must show $n \in B$. By (PL5), either $n = 0$ or $n = S(k)$ for some k .

Case $n = 0$. By the left side of (1), $n = 0 \in B$.

Case $n = S(k)$ for some $k \in \mathbb{N}$. Then $k \in {}_S[n]$, and ${}_S[n] \subseteq B$ gives $k \in B$. By the right side of (1), $k+1 \in B$. But $k+1 = S(k) = n$, so $n \in B$.

Secondly, $\varphi_2(B) \rightarrow \varphi_1(B)$. Assume

$$\forall n \in \mathbb{N}. ({}_S[n] \subseteq B \rightarrow n \in B). \quad (2)$$

W.T.S

$$0 \in B \wedge \forall n \in \mathbb{N}. (n \in B \rightarrow n+1 \in B).$$

$0 \in B$?

By (PL4), no k has $S(k) = 0$, so ${}_S[0] = \emptyset \subseteq B$. By (2), $0 \in B$.

$\forall n \in \mathbb{N}. (n \in B \rightarrow n+1 \in B)$?

Take arbitrary $n \in \mathbb{N}$ with $n \in B$. We want $n+1 \in B$, and by (2) applied to $n+1$ it suffices to show ${}_S[n+1] \subseteq B$. Take any $k \in {}_S[n+1]$. Then $S(k) = n+1 = S(n)$, so $k = n$ by (PL3). But $n \in B$, so $k \in B$. Hence ${}_S[n+1] \subseteq B$.

So $\varphi_1(B) \longleftrightarrow \varphi_2(B)$ for every $B \subseteq \mathbb{N}$, and Lemma 1.6 gives the theorem. $\square$

¹W.T.S := Want to Show